

SHOPEE BARCODE & VIDEO MANAGEMENT SYSTEM

Use Case Specification Document

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Document	Use Case Specification	Version	1.0
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1. Introduction

1.1 Purpose

This document provides detailed use case specifications for the Shopee Barcode & Video Management System. It describes all interactions between users (employees and administrators) and the system, serving as the primary behavioral specification for the development team and QA engineers.

1.2 Actor Definitions

Actor	Description
Employee	Warehouse or store staff who perform daily order scanning, video recording, and complaint submission. Primary user of the mobile application.
Admin	System administrator with full access to all features including staff, store, video, and complaint management.
System	The backend system that performs automated actions such as token validation, file upload processing, and backend auto-discovery.

1.3 Use Case Inventory

UC ID	Use Case Name	Primary Actor	Priority
UC-01	Register Account	Employee / Admin	High
UC-02	Login to System	Employee / Admin	High
UC-03	Scan Order QR Code	Employee	High
UC-04	Record and Upload Video	Employee	High
UC-05	View Video History	Employee	Medium
UC-06	Submit Complaint	Employee	Medium
UC-07	Update Profile	Employee / Admin	Low
UC-08	Change Password	Employee / Admin	Medium
UC-09	Configure App Settings	Employee / Admin	Low
UC-10	Manage Staff Accounts	Admin	High
UC-11	Manage Stores	Admin	Medium
UC-12	Manage Video Records	Admin	High
UC-13	Manage and Resolve Complaints	Admin	High

2. Detailed Use Cases

UC-01: Register Account

Use Case ID	UC-01
Use Case Name	Register Account
Actors	Employee, Admin
Preconditions	User is not logged in; has access to the registration screen.
Postconditions	New user account is created in the database. User can now log in.
Priority	High

Main Flow:

1. User opens the application and navigates to the Register screen.
2. User enters full name, email address, password, and selects role (default: Employee).
3. User submits the registration form.
4. System validates all fields: name is non-empty, email matches valid format, password meets minimum length.
5. System checks the database for duplicate email address.
6. System creates a new user account and stores the hashed password.
7. System returns the new user object; application displays success message.
8. User is redirected to the Login screen.

Alternative Flows:

- A-1 Admin Registration: If the user selects the 'Admin' role, the account is created with admin privileges. All steps remain the same.

Exception Flows:

- E-1 Duplicate Email: If the email already exists, system returns HTTP 400 and displays 'Email already registered' error.
- E-2 Invalid Input: If any field fails validation, the form displays inline error messages and does not submit.

UC-02: Login to System

Use Case ID	UC-02
Use Case Name	Login to System
Actors	Employee, Admin
Preconditions	User has a registered account; application is on the Login screen.
Postconditions	User is authenticated; JWT token is stored locally. User is on the Home screen.

Priority

High

Main Flow:

9. User enters their registered email address and password.
10. User taps the 'Login' button.
11. System validates credentials against the database (bcrypt hash comparison).
12. System generates a JWT access token.
13. System returns the JWT token and user profile (user_id, name, email, role).
14. Application stores the token and user data locally.
15. User is redirected to the Home screen. Admin users see additional Admin Dashboard option.

Alternative Flows:

- A-1 Admin Login: Admin users are redirected to the same Home screen but have access to the Admin Dashboard tab.

Exception Flows:

- E-1 Invalid Credentials: System returns HTTP 400; application displays 'Invalid credentials' alert.
- E-2 Empty Fields: Application displays validation error; login request is not submitted.

UC-03: Scan Order QR Code

Use Case ID	UC-03
Use Case Name	Scan Order QR Code
Actors	Employee
Preconditions	Employee is logged in; is on the Scan screen; camera and microphone permissions have been granted.
Postconditions	Order ID is captured and displayed. System is ready for video recording (UC-04).
Priority	High

Main Flow:

16. Employee navigates to the Scan screen from the bottom navigation bar.
17. System requests camera and microphone permissions if not already granted.
18. System activates the rear camera in QR scanning mode.
19. Employee points the camera at the Shopee order package QR code / barcode.
20. System detects the QR code and decodes the order ID string.
21. System displays the decoded order ID on the scan overlay.
22. The 'Start Recording' button becomes enabled.

Alternative Flows:

- A-1 Multiple Scans: If the employee scans a new QR code while the previous one is displayed, the order ID is updated to the newest scan. Recording must not be active.
- A-2 Re-scan: Employee can re-scan at any time before recording starts to update the order ID.

Exception Flows:

- E-1 Permission Denied: If camera or microphone permission is denied, system displays an appropriate message and the Scan screen is disabled.
- E-2 Unreadable QR: If the QR code is damaged or unreadable, no order ID is captured. Employee must attempt re-scan.

UC-04: Record and Upload Video

Use Case ID	UC-04
Use Case Name	Record and Upload Video
Actors	Employee
Preconditions	UC-03 is completed; order ID is displayed; camera is ready; employee is on the Scan screen.
Postconditions	Video file is stored on the server. Video metadata record (video_id, order_id, user_id, timestamps) is saved in the database.
Priority	High

Main Flow:

23. Employee taps 'Start Recording'.
24. System begins video recording via the device camera.
25. System displays a timer showing elapsed recording duration in seconds.
26. Employee performs the packing/handling action while recording.
27. Employee taps 'Stop Recording'.
28. System stops the recording and saves the video file locally (MP4 format).
29. System uploads the video file to the backend using multipart form data, including order_id, user_id, and start_time.
30. Backend saves the video file to the server and creates a video metadata record in the database.
31. System displays a success alert: 'Upload video successful!'
32. Employee is redirected to the Home screen.

Alternative Flows:

- A-1 Long Recording: There is no enforced maximum recording duration. Employee may record as long as needed.

Exception Flows:

- E-1 Upload Failure: If network fails or timeout (60s) is exceeded, system displays 'Upload error' alert with error details. Video file remains on device.
- E-2 Camera Not Ready: If camera is not initialized when start is tapped, system displays 'Camera not ready' error.

- E-3 No Order ID: If start is tapped without a scanned QR code, system displays 'Please scan QR first' error.

UC-05: View Video History

Use Case ID	UC-05
Use Case Name	View Video History
Actors	Employee
Preconditions	Employee is logged in; JWT token is valid.
Postconditions	Employee can see all their video records.
Priority	Medium

Main Flow:

33. Employee navigates to the History screen from the bottom navigation bar.
34. System sends a GET request to /videos/me with the employee's JWT token.
35. Backend validates the token, retrieves all videos associated with the employee's user_id.
36. System displays the list of videos with order ID, file name, and recording timestamps.

Alternative Flows:

- A-1 No History: If the employee has no recorded videos, the screen displays an empty state message.

Exception Flows:

- E-1 Token Expired: If JWT is invalid or expired, system returns HTTP 401 and redirects user to Login screen.
- E-2 Network Error: System displays a generic error message; history list is empty.

UC-06: Submit Complaint

Use Case ID	UC-06
Use Case Name	Submit Complaint
Actors	Employee
Preconditions	Employee is logged in; order ID is known.
Postconditions	Complaint record is saved with resolved=false. Admin can view and resolve the complaint.
Priority	Medium

Main Flow:

37. Employee navigates to the complaint submission screen.

38. Employee enters the order ID, selects an issue type (Missing Item / Wrong Item / Other), and provides a description.
39. Employee submits the complaint form.
40. System validates input fields.
41. System verifies the order exists in the database.
42. System creates a new complaint record with status 'unresolved'.
43. System returns the complaint details; application displays success confirmation.

Alternative Flows:**Exception Flows:**

- E-1 Order Not Found: If the provided order_id does not exist, system returns HTTP 404 and displays 'Order not found' error.
- E-2 Invalid Issue Type: System rejects any issue_type not in the defined enum (missing_item, wrong_item, other).

UC-07: Update Profile

Use Case ID	UC-07
Use Case Name	Update Profile
Actors	Employee, Admin
Preconditions	User is logged in.
Postconditions	User profile is updated in the database.
Priority	Low

Main Flow:

44. User navigates to the Profile screen.
45. System displays current user information (name, email, store details).
46. User edits their name and/or email address.
47. User submits the update form.
48. System validates input (email uniqueness check).
49. System updates the user record and returns updated user data.
50. Application displays updated profile information.

Alternative Flows:

Exception Flows:

- E-1 Duplicate Email: If the new email is already in use by another account, system returns HTTP 400.
- E-2 User Not Found: If user_id is invalid, system returns HTTP 404.

UC-08: Change Password

Use Case ID	UC-08
Use Case Name	Change Password
Actors	Employee, Admin
Preconditions	User is logged in.
Postconditions	User's password is updated. Old password is no longer valid.
Priority	Medium

Main Flow:

51. User navigates to the Change Password screen.
52. User enters their current (old) password and the new password.
53. User submits the form.
54. System verifies the old password against the stored hash.
55. System hashes the new password and updates the user record.

56. System returns success; application displays confirmation message.

Alternative Flows:

Exception Flows:

- E-1 Wrong Old Password: System returns HTTP 400 with 'Old password is incorrect' message.
- E-2 Same Password: Application-level validation prevents reuse of the same password (recommended enhancement).

UC-09: Configure App Settings

Use Case ID	UC-09
Use Case Name	Configure App Settings
Actors	Employee, Admin
Preconditions	User is logged in.
Postconditions	User preferences are saved. UI reflects the selected theme and language.
Priority	Low

Main Flow:

57. User navigates to the Settings screen.
58. System loads current settings (dark_mode, language) from the backend.
59. User toggles dark mode on/off and/or changes language preference.
60. System saves updated settings to the backend.
61. Application immediately applies the new theme/language.

Alternative Flows:

Exception Flows:

- E-1 Network Failure: If settings cannot be saved, local settings are applied temporarily but a sync error is noted.

3. Admin Use Cases

UC-10: Manage Staff Accounts

Use Case ID	UC-10
Use Case Name	Manage Staff Accounts
Actors	Admin
Preconditions	Admin is logged in and on the Admin Dashboard.
Postconditions	Selected staff account is permanently removed from the system.
Priority	High

Main Flow:

62. Admin navigates to the Manage Staff section of the Admin Dashboard.
63. System retrieves all users with role 'employee' from the backend.
64. System displays a list of staff accounts with name, email, and role.
65. Admin selects a staff account to delete.
66. System displays a confirmation dialog.
67. Admin confirms the deletion.
68. System sends DELETE /staff/{user_id} request.
69. Backend deletes the user record. System refreshes the staff list.

Alternative Flows:

- A-1 View Only: Admin may browse staff accounts without performing any action.

Exception Flows:

- E-1 User Not Found: If the user_id is invalid, system returns HTTP 404 and displays an error.
- E-2 Cancel Deletion: Admin selects 'Cancel' on the confirmation dialog; no action is taken.

UC-11: Manage Stores

Use Case ID	UC-11
Use Case Name	Manage Stores
Actors	Admin
Preconditions	Admin is logged in.
Postconditions	Store record is updated or deleted as requested.
Priority	Medium

Main Flow:

70. Admin navigates to the Manage Stores section.

71. System retrieves all store records from GET /store.
72. System displays stores with name, email, phone, and location.
73. Admin selects a store to edit or delete.
74. For edit: Admin updates store fields and submits. System calls PUT /store.
75. For delete: Admin confirms deletion. System calls DELETE /store/{store_id}.
76. System refreshes the store list.

Alternative Flows:**Exception Flows:**

- E-1 Store Not Found: If store_id is invalid, system returns HTTP 404.

UC-12: Manage Video Records

Use Case ID	UC-12
Use Case Name	Manage Video Records
Actors	Admin
Preconditions	Admin is logged in.
Postconditions	Video metadata is removed from the database. Physical file is deleted if it exists.
Priority	High

Main Flow:

77. Admin navigates to the Manage Videos section.
78. System retrieves all video records from GET /videos.
79. System displays a list of all videos with order ID, file name, uploader, and timestamps.
80. Admin selects a video record to delete.
81. Admin confirms the deletion.
82. System sends DELETE /videos/{video_id}.
83. Backend deletes the video metadata record from the database.
84. Backend attempts to delete the physical video file from the server filesystem.
85. System refreshes the video list.

Alternative Flows:

- A-1 File Not Found: If the physical file does not exist on disk (already manually deleted), the metadata record is still removed from the database and no error is raised to the user.

Exception Flows:

- E-1 Video Not Found: If video_id is invalid, system returns HTTP 404.

UC-13: Manage and Resolve Complaints

Use Case ID	UC-13
Use Case Name	Manage and Resolve Complaints
Actors	Admin
Preconditions	Admin is logged in.
Postconditions	Complaint is marked as resolved with timestamp. Alternatively, complaint record is deleted.
Priority	High

Main Flow:

86. Admin navigates to the Manage Complaints section.
87. System retrieves all complaint records from GET /complaints.
88. System displays complaints with order ID, issue type, description, and resolution status.
89. Admin reviews a complaint.
90. Admin selects 'Resolve' for an unresolved complaint.
91. System sends PATCH /complaints/{complaint_id}/resolve.
92. Backend sets resolved=true and records resolved_at timestamp.
93. System refreshes the complaint list.
94. Admin may also delete complaint records using DELETE /complaints/{complaint_id}.

Alternative Flows:

- A-1 Already Resolved: If admin attempts to resolve an already-resolved complaint, system returns HTTP 400 with 'Complaint is already resolved'.

Exception Flows:

- E-1 Complaint Not Found: If complaint_id is invalid, system returns HTTP 404.

4. Use Case Relationships

4.1 Include Relationships

Base Use Case	Includes	Description
UC-04: Record Video	UC-03: Scan QR Code	Video recording requires a prior QR scan
All Authenticated UCs	UC-02: Login	All post-login features require a valid session

4.2 Extend Relationships

Base Use Case	Extended By	Condition
UC-02: Login	Admin Dashboard Access	Extends when role = admin
UC-12: Manage Videos	Physical File Deletion	Extends when storage_url file exists on disk